

Mahony Filter:

- **Mahony/Medwick** – these filters are the most computationally efficient since they usually require few flops per computation since it doesn't involve any matrix operations nor Jacobians & fits into light embedded programs.
 - Propagates the full state forward (quaternion integration from gyro)
 - Computes a error signal by comparing a model prediction
 - Feeds the error back into the propagation

2D rotation example:

Take these 2 vectors R^2 : $a(\cos(a), \sin(a))$, $p(\cos(p), \sin(p))$

If $a = 45$ deg & $b = 135$

Taking the cross product:

$$e_z = a_x p_y - p_x a_y = \cos(45) \sin(135) - \cos(135) \sin(45) = 1$$

$$\sin^{-1}(e) = 90$$

For normalized vectors separated by 90 degrees, the cross product error has magnitude of 1 and points along the rotation axis describing their mismatch. The error is represented by $\sin(90)$, not a angular rate.

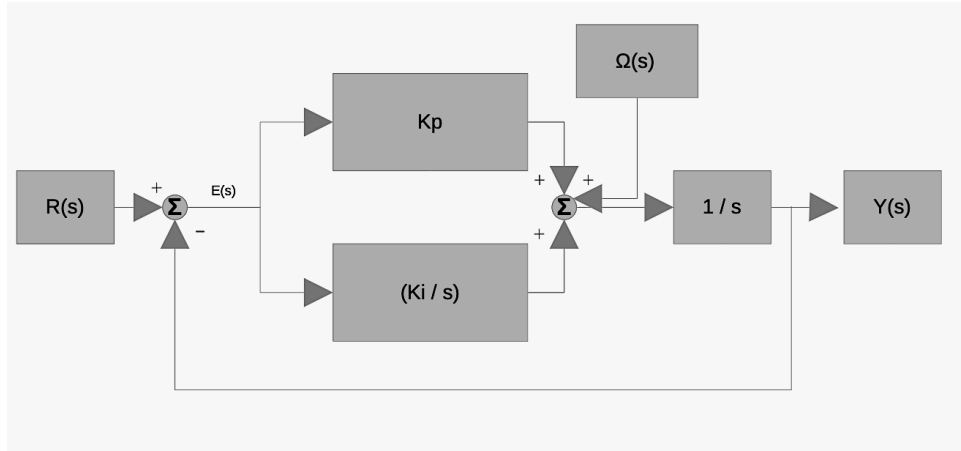
KI

The integral term multiplies this error by K_i and accumulates it into an estimated gyro – rate correction, where a persistent one signed error causes the bias estimate to move until it cancels the persistent gyro offset. When the estimate overshoots the error changes sign and the bias estimate begins moving back. Integral action alone is delayed and undamped so it tends to oscillate.

KP

The proportional term multiplies the current error by K_p and adds the result directly to the gyro rate. It provides a immediate correction toward the reference. When the error reverses during overshoot, the Proportional correction reverses immediately, damping the oscillation. The bias state remains available to cancel persistent gyro offset after the cross product error returns to zero.

Small angle linearized Mahony model:



$$Y(s) = \frac{K_p s + K_i}{s^2 + K_p s + K_i} R(s) + \frac{s}{s^2 + K_p s + K_i} \Omega(s)$$

$Y(s)$: estimated attitude angle

$R(s)$: accelerometer – derived attitude reference

$\Omega(s)$: Measured gyro angular rate, $\frac{\text{rad}}{\text{s}}$

$$D(s) = s^2 + K_p s + K_i$$